

5513assignment 1

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Introduction

This report examines the relationship between marketing spending and customer purchases. I want to find out if higher marketing expenditure leads to more customer spending. This data comes from a personalized marketing campaign in the past quarter. The analysis will help businesses create more effective marketing strategies.

Research Question

The core research question of this report is: Does higher marketing spend lead to higher purchase amount for customers? I will analyze the correlation between marketing investment and customer spending behavior using this dataset. I will also further examine whether there are differences in marketing effectiveness across different customer segments.

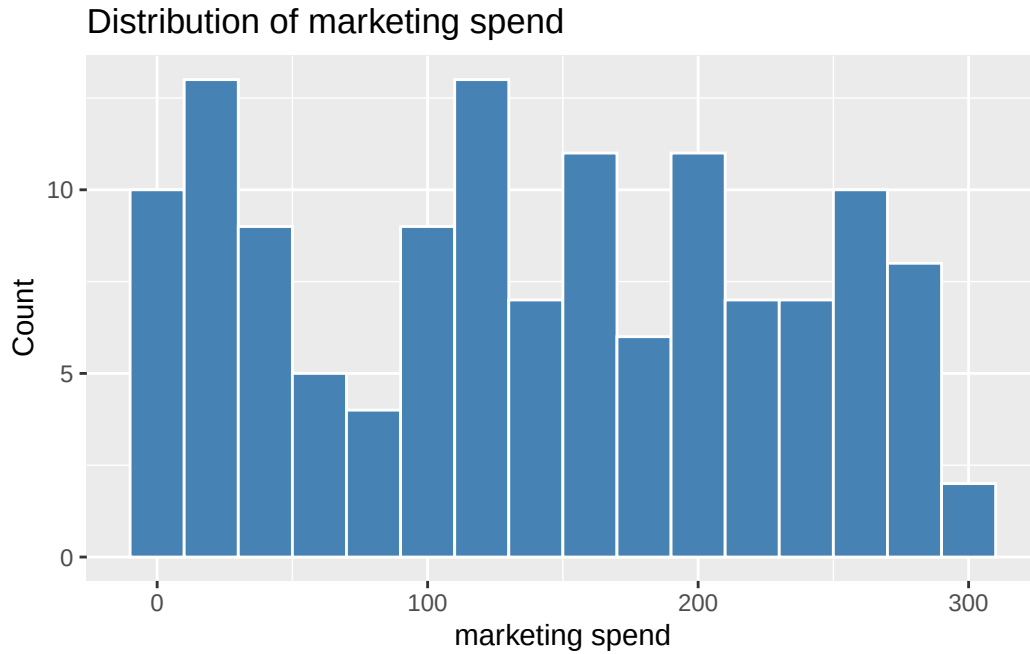
Dataset Introduction

This dataset comes from the personalized marketing records of an online retailer, with 150 customer samples. It covers key fields like customer ID, signup date, marketing spend, visits, purchase amount, and customer segment. Missing values in the purchase amount variable were removed during data cleaning. The cleaned dataset is neat and complete, which is enough for us to analyze the relationship between marketing investment and consumption performance.

[1] 132 7

```
[1] "customer_id"      "signup_date"      "marketing_spend" "visits"
[5] "purchase_amount" "segment"          "notes"
```

The data contains 132 rows and 7 columns. And it contains customer_id, signup_date, marketing_spend, visits, purchase_amount, segment, notes 7 variables. The cleaned dataset retains 132 valid observations with complete variable counts, which can be directly used for subsequent statistical analysis and data visualization. All variables have undergone data type verification and missing value cleaning. Specifically, marketing_spend (marketing spend) and purchase_amount (purchase amount) are numerical variables, while segment (customer segment) is a character variable.



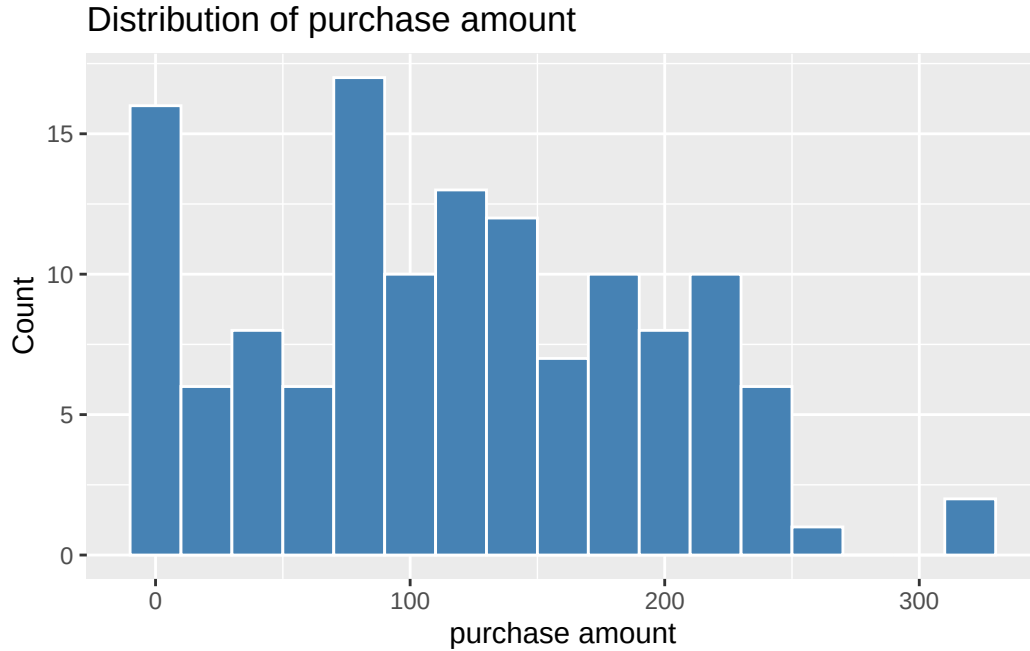


Figure 1 shows the histogram of marketing spend, and Figure 2 shows the histogram of purchase amount. Both figures indicate that the values of the two variables are concentrated in the 0-200 range, with an overall right-skewed distribution. This suggests that most customers have moderate marketing investment and purchase amounts, while a small number of customers have high investment and consumption, providing a data background for the subsequent correlation analysis.

Table 1: Table:Describe the statistical characteristics of the two variables

marketing_spend	purchase_amount	visits
Min. : 0.29	Min. : 0.00	Min. : 0.000
1st Qu.: 55.94	1st Qu.: 58.15	1st Qu.: 4.000
Median :133.61	Median :112.42	Median : 5.000
Mean :139.42	Mean :116.32	Mean : 5.553
3rd Qu.:211.03	3rd Qu.:173.75	3rd Qu.: 7.000
Max. :295.57	Max. :313.34	Max. :18.000

Table lists the statistical summary of these two variables, including the minimum, mean, and maximum values. As shown in the table, the average marketing spend is 139.42, and the average purchase amount is 116.32. These figures provide a basic background for the analysis.

Result

Call:

```
lm(formula = purchase_amount ~ marketing_spend, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-114.816	-32.624	-7.954	32.163	154.167

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	23.76634	7.89505	3.01	0.00314 **
marketing_spend	0.66382	0.04777	13.89	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.7 on 130 degrees of freedom

Multiple R-squared: 0.5976, Adjusted R-squared: 0.5945

F-statistic: 193.1 on 1 and 130 DF, p-value: < 2.2e-16

The results of this study clearly answer the core research question: higher marketing spend leads to higher purchase amount for customers.

Figure 1, a scatter plot with a fitted line, visually shows a clear positive correlation between marketing spend and purchase amount. Linear regression results confirm this: for every 1-unit increase in marketing spend, purchase amount increases by an average of 0.66, and this result is highly statistically significant ($p < 0.001$). The model explains nearly 60% of the variation in purchase amount.

Figure 2, the boxplot further shows that across different customer segments, groups with higher marketing spend also have generally higher purchase amounts, and this pattern holds for all customer groups.

Overall, the data fully proves that increasing marketing investment in customers can effectively improve their purchase amount.

The Relationship Between marketing spend per customer and

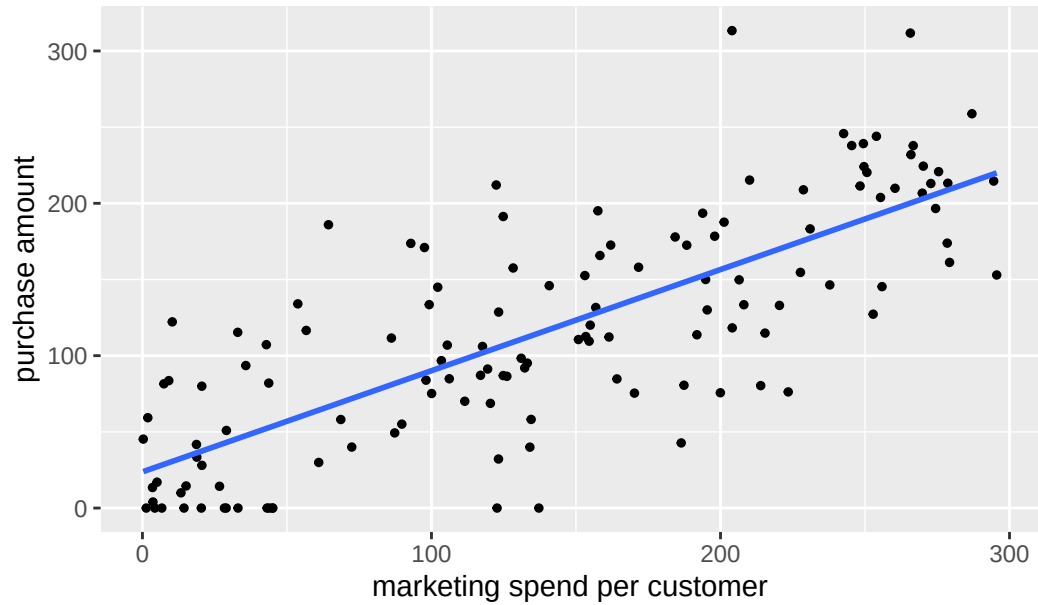


Figure 1: marketing spend per customer VS purchase amount

The Relationship Between marketing spend per customer and

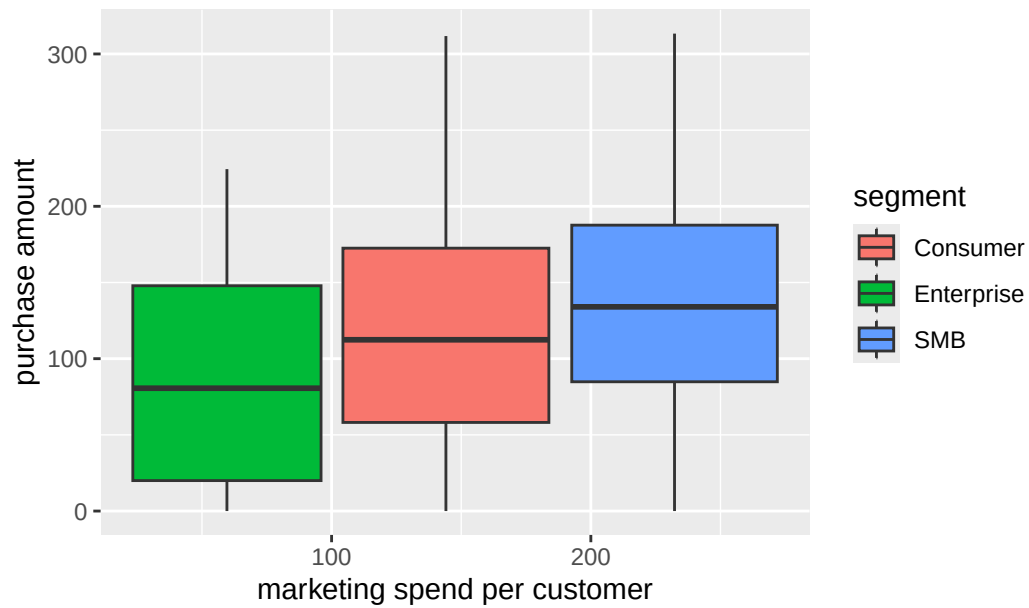


Figure 2: marketing spend per customer VS purchase amount